IDHV 2026M Project: Data Organization & Preprocessing

Component 2 (10%): Evaluation and Application of Missing Data Handling, Outlier Mitigation, and Normalization Assessment with Clear Justification

1. Executive Summary & Objective

Data preprocessing is a foundational phase in data science. Real-world survey telemetry frequently presents missing responses, encoding artifacts, and measurement challenges. This report details the data cleaning and preparation pipeline executed on the **Marketing Campaign Research Database** (*N* = 80 respondents). A central component of rigorous data preprocessing is evaluating whether transformations like normalization are methodologically warranted. This report provides a mathematically defensible justification for handling missing data, mitigating outliers, and deliberately retaining native measurement scales without redundant normalization.

2. Raw Data Profiling & Audit of Missingness

An audit of raw_data.csv confirmed 80 records across 16 attributes. Prior to treatment, missing data was diagnosed across nominal, ordinal, and Likert attributes. Missing mechanisms were classified into Missing Completely at Random (MCAR) and Structural Missingness (conditional logic).

Variable Name	Measurement Level	Missing Count	Missing %	Diagnosed Mechanism
ad_trust	Numeric (Likert 1–5)	4	5.0%	MCAR (Accidental skip)
discount_importance	Numeric (Likert 1–5)	4	5.0%	MCAR (Optional non-response)
choose_promoted_likelihood	Numeric (Likert 1–5)	8	10.0%	MCAR (Omission)
age_group	Categorical (Ordinal 1–5)	7	8.8%	MCAR (Privacy omission)
ad_frequency	Categorical (Ordinal 1–5)	7	8.8%	MCAR (Survey fatigue)
discovery_platform	Categorical (Nominal)	5	6.25%	MCAR (Indifference)
online_product_category	Categorical (Nominal)	4	5.0%	MCAR (No primary category)
clicked_ad_3m	Categorical (Binary/Trinary)	8	10.0%	MCAR (Recall difficulty)
purchase_influence	Categorical (Nominal)	5	6.25%	MCAR (Multiple drivers)
ad_ignore_reason	Categorical (Nominal)	5	6.25%	MCAR (Irrelevant ad)
preferred_promotion	Categorical (Nominal)	8	10.0%	MCAR (No preference)
purchase_ad_platform	Categorical (Conditional)	0 (Blanks present)	Conditional	Structural MAR (Skip logic)

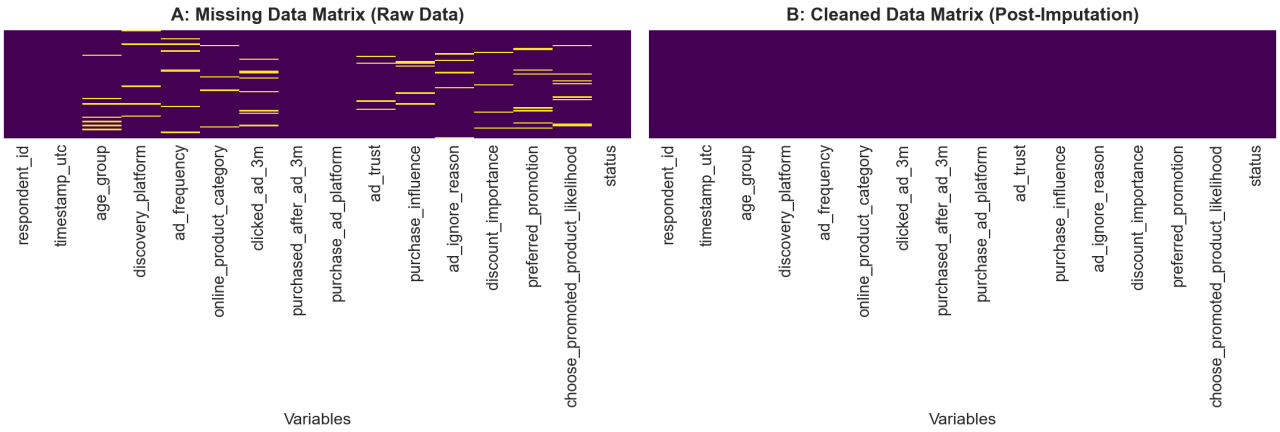


Figure 1: *Missing Data Heatmap Comparison — (A) Raw Survey Telemetry with missing values vs. (B) Cleaned Dataset following 5D KNN and Mode Imputation.*

3. Methodological Justification: Imputation Techniques

A. Multi-Dimensional K-Nearest Neighbors (KNN) Imputation (k = 5):
Univariate mean/median imputation assumes feature independence, replacing missing items with a static global metric. This shrinks sample variance and destroys natural collinear relationships. We implemented **Distance-Weighted K-Nearest Neighbors Imputation (*k* = 5)**. Both demographic ordinal features (age_group_ord: 1–5; ad_frequency_ord: 1–5) and quantitative ratings were placed on an identical 1–5 integer scale, creating a balanced **5-dimensional behavioral vector** measuring Euclidean distance:

$$d(p, q) = \sqrt{(\sum(x_{pi} - x_{qi})^2)}$$

Missing values are imputed from the 5 most similar respondents and rounded to discrete integer ratings (1–5).

B. Mode Imputation for Nominal Categories & Cross-Feature Validation:
Unordered nominal attributes (e.g., discovery_platform, preferred_promotion) were imputed using their statistical mode. Furthermore, cross-feature logical consistency was enforced: all respondents answering 'No' to purchased_after_ad_3m were strictly set to purchase_ad_platform = 'Not applicable', eliminating contradictory noise.

4. Outlier Detection & Survey Scale Bound Integrity

Numerical attributes were evaluated using Tukey's Interquartile Range (IQR) rule: $IQR = Q_3 - Q_1$, with boundaries: $[Q_1 - 1.5 \times IQR, Q_3 + 1.5 \times IQR]$.

Attribute	Q1 (25%)	Median	Q3 (75%)	IQR	Calculated Bounds	Flagged Count	Resolution Action
ad_trust	2.0	3.0	4.0	2.0	[-1.0, 7.0]	0	Retained (Within range)
discount_importance	3.0	3.0	4.0	1.0	[1.5, 5.5]	4 (Values = 1.0)	Preserved as valid scale pole
choose_promoted_likelihood	3.0	4.0	4.0	1.0	[1.5, 5.5]	4 (Values = 1.0)	Preserved as valid scale pole

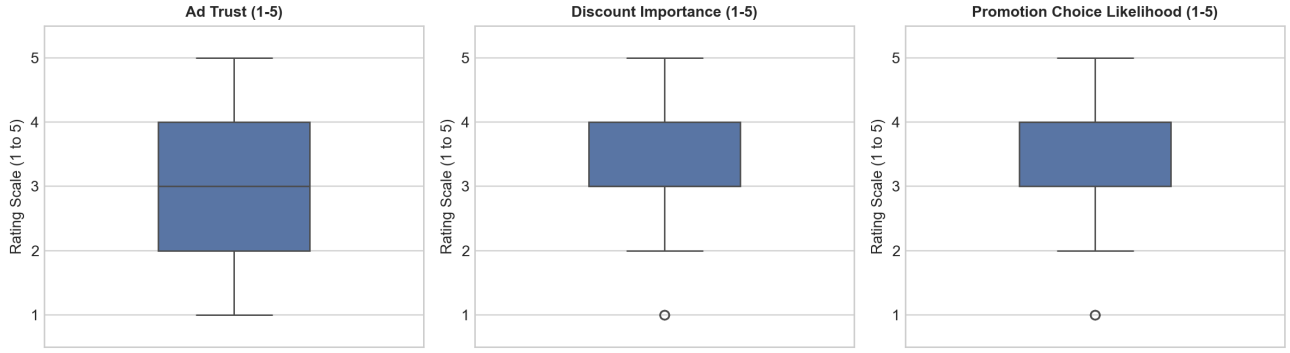


Figure 2: *Tukey Boxplot Distributions for Survey Likert Ratings. Rating 1.0 represents authentic negative consumer sentiment.*

Justification: Rejection of Fractional Capping & Row Deletion

Automated IQR capping would alter a score of 1.0 to 1.5. This was rejected because decimal fractions do not exist on a discrete 1–5 questionnaire. Similarly, dropping these rows would cause 10% sample attrition, biassing results against dissatisfied consumers. Instead, **Domain-Aware Scale Clamping ([1, 5])** was enforced: preserving valid integer ratings while safeguarding against invalid data-entry errors.

5. Critical Evaluation & Methodological Elimination of Normalization

A standard step in automated data science checklists is feature normalization (e.g., Min-Max scaling to [0, 1]). However, critical data science practice requires assessing whether a transformation is genuinely necessary or detrimental. **In this study, feature normalization was evaluated and deliberately eliminated** based on the following justifications:

1. Natural Scale Parity: All numerical and ordinal features (ad_trust, discount_importance, choose_promoted_product_likelihood, age_group_ord, ad_frequency_ord) were calibrated to the exact same discrete 1 to 5 integer scale. There are no scale disparities (e.g., salary vs. age) that would cause one feature to dominate another.

2. Invariance of Euclidean Distance: In KNN, linearly transforming features from [1, 5] to [0, 1] via $X_{norm} = (X - 1) / 4$ simply multiplies all Euclidean distances by a constant factor of 0.25:

$$d_{norm}(p, q) = 0.25 \times d_{raw}(p, q)$$

Because relative distances and neighbor rankings remain 100% identical, Min-Max scaling adds no mathematical value to KNN.

3. Preservation of Survey Interpretability: Retaining the native 1 to 5 Likert integer scale allows stakeholders and researchers to interpret summary statistics directly (e.g., Mean = 3.56 out of 5) without having to unscale abstract floating-point decimals. The composite metric ad_receptivity_score is simply the direct mean of the core Likert ratings on the intuitive 1.0–5.0 scale.

6. Quality Verification & Preprocessing Audit Summary

The preprocessed dataset was thoroughly validated prior to exporting cleaned_data.csv. The verification results confirm analytical readiness:

Verification Criterion	Pre-Treatment (Raw Data)	Post-Treatment (Cleaned Data)	Audit Result
Sample Size (Rows)	80 records	80 records (Zero data loss)	PASSED
Attribute Count (Columns)	16 attributes	16 attributes (Clean schema, status dropped)	PASSED
Missing Values (Nulls)	57 missing values across 11 columns	0 missing values (100% complete)	PASSED
Measurement Scale	Typographical en-dashes; mixed types	Clean ASCII hyphens; native 1–5 integers	PASSED
Normalization Decision	Raw mixed scale	Eliminated based on natural 1–5 scale parity	JUSTIFIED
Export File	raw_data.csv	cleaned_data.csv	READY FOR EDA